

Gashahun Demise

Machine Learning Engineer | Computer Vision Engineer | AI Engineer

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Professional Summary

Machine Learning and Computer Vision Engineer with experience in computer vision, deep learning, dataset engineering, model evaluation, and software development. Skilled in Python, TensorFlow, OpenCV, and Full-stack development technologies, with experience building practical AI systems from data preparation through edge deployment.

Technical Skills

Machine Learning & Computer Vision: TensorFlow, Keras, scikit-learn, OpenCV, deep learning, transfer learning, image classification, object detection, segmentation

Programming & Data: Python, C/C++, SQL, JavaScript/TypeScript, NumPy, Pandas

Software Engineering: FastAPI, Django, React, REST APIs, PostgreSQL, MySQL

Tools & Deployment: TensorFlow Lite, Docker, Linux, Git/GitHub, CI/CD

Professional Experience

Computer Vision Researcher

Jul 2025 – Present

Ethiopian Artificial Intelligence Institute (EAI)

Addis Ababa, Ethiopia

- Research and develop computer vision systems for plant disease classification and crop and animal growth monitoring.
- Build and manage image annotation pipelines for classification and segmentation.
- Define class taxonomies, labeling guidelines, edge-case rules, and QA procedures to improve annotation consistency and dataset quality.
- Perform dataset preprocessing, label correction, and format conversion across YOLO, COCO, and Pascal VOC to support reproducible model development and evaluation.

Data Communication and Security Intern

2024 - 2025

AASTU ICT and Research Center

Addis Ababa, Ethiopia

- Supported university-wide data communication and network-security operations through infrastructure monitoring and troubleshooting.
- Documented operational and security procedures and assisted with diagnosis of connectivity and system issues across the university ICT environment.

Selected Projects

AgriScan - Enset Plant Disease Detection System

2025 - Present

Python, TensorFlow/Keras, OpenCV, TensorFlow Lite

- Collected, labeled, cleaned, and validated 7,545 Enset leaf images across healthy, bacterial-wilt, and naturally aged classes.
- Trained and compared five deep learning models using transfer learning and fine-tuning, achieving up to 95.42% accuracy; a custom model reached 97.5%.
- Evaluated performance using five-fold cross-validation, precision, recall, F1-score, confusion matrices, and Grad-CAM.
- Optimized the final model with quantization and TensorFlow Lite for mobile and edge deployment.

Hope Lounge - QR-Based Food Ordering Platform

Dec 2025 – Apr 2026

React, Django, PostgreSQL, REST APIs

- Led development of a QR-based ordering platform from design through delivery, integrating table-side menu browsing, order placement, status tracking, and kitchen visibility.
- Digitized the restaurant's table-ordering workflow, reducing manual order-taking staffing needs by 50% and improving operational visibility.

Student Internship Monitoring System

Oct 2024 - Jun 2025

React, Django, PostgreSQL

- Led system design and frontend development of a multi-role platform consolidating progress tracking, daily logs, grading, and evaluation reporting.
- Built chatbot and ratings-based recommendation components to automate FAQ handling and support student-company matching.

Education

B.Sc. in Electrical and Computer Engineering

Jul 2021 - Jun 2025

Addis Ababa Science and Technology University (AASTU)

Addis Ababa, Ethiopia

GPA: 3.45/4.00 | National Exit Examination: 71%

Certification & Languages

Certification: Deep Learning for Computer Vision - WorldQuant University

Languages: Afaan Oromoo (Native), Amharic (Native), English (Professional Working)